

Problem Statement

In today's highly competitive telecommunications industry, retaining existing customers is significantly more cost-effective than acquiring new ones. However, telecom companies often struggle to identify customers who are likely to discontinue their services due to factors such as high service charges, frequent customer support interactions, excessive roaming costs, contract expiration, or dissatisfaction with available plans. As a result, businesses experience increased customer churn, leading to revenue loss, reduced customer lifetime value, and higher marketing expenditures.

Traditional methods of identifying at-risk customers rely heavily on manual analysis and historical observations, which are time-consuming, inefficient, and often unable to detect churn patterns early enough for effective intervention. With the growing volume of customer data, there is a need for an intelligent and automated solution capable of analyzing customer behavior and predicting the likelihood of churn in real time.

This project aims to develop an **AI-Powered Customer Churn Prediction & Retention Analytics Platform** using **IBM Watson Machine Learning and AutoAI**. The system leverages machine learning models trained on customer behavioral data, including account tenure, monthly charges, data usage, customer service interactions, contract renewal status, and roaming activity, to predict whether a customer is likely to churn.

The proposed solution provides a user-friendly web interface through which telecom service providers can input customer information and instantly receive churn risk predictions. By identifying high-risk customers before they leave, organizations can implement targeted retention strategies, improve customer satisfaction, reduce revenue loss, and make data-driven business decisions.

The project demonstrates the practical application of Artificial Intelligence, Machine Learning, IBM Cloud services, and web technologies to solve a real-world business problem while enabling proactive customer retention and operational efficiency.

Objective

To design and develop a web-based intelligent churn prediction system that utilizes **IBM Watson Machine Learning AutoAI models** to analyze customer behavior and predict churn risk, enabling businesses to take preventive actions and improve customer retention rates.